A PROOF OF CALDERON'S MOD-p BAILEY CONGRUENCE AT EVERY ODD INERT PRIME

Abstract. Let ω be a root of $x^2 - Tx + N$ with $T, N \in \mathbb{Z}$, let $\mathcal{O} = \mathbb{Z}[\omega]$, and for $A \geq C \geq 1$, $B \geq D \geq 1$ let $R_\omega(A, B; C, D)$ be the rectangular coefficient of Calderon [arXiv:2608.00347v1]. His Conjecture 6.5 asserts a mod-p digit factorisation of R_ω for every imaginary quadratic order and every prime satisfying his hypothesis (6.3). We prove it, under the weaker hypothesis that p is odd, $p \nmid \Delta := T^2 - 4N$ and $\left(\frac{\Delta}{p}\right) = -1$; neither $p > 5$ nor $\Delta < 0$ is used. The proof is elementary: the only inputs are that $\{1, \omega\}$ is an $\mathbb{F}_p$-basis of $\mathcal{O}/p\mathcal{O} \cong \mathbb{F}_{p^2}$ and that $\omega^p \equiv \bar{\omega}$ there. The mechanism is the one the author sketches in the twenty-five lines above his conjecture; what this note supplies is the verification he defers.

1. The conjecture

Fix $T, N \in \mathbb{Z}$ and let ω be a root of $x^2 - Tx + N$ with conjugate $\bar{\omega} = T - \omega$ and discriminant $\Delta = T^2 - 4N$. Put $\mathcal{O} = \mathbb{Z}[\omega]$, and for a finite box $\mathcal{B} \subset \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 1}$ put

$$F(\mathcal{B}) = \prod_{(u,v) \in \mathcal{B}} (u + v\omega) \in \mathcal{O}.$$

For $A \geq C \geq 1$ and $B \geq D \geq 1$ the *rectangular coefficient* is

$$R_\omega(A, B; C, D) = \frac{F\big([A - C + 1, A] \times [B - D + 1, B]\big)}{F\big([1, C] \times [1, D]\big)} \in \mathbb{Q}(\omega). \tag{1}$$

For a prime p write $\mathcal{O}_{\omega,p} = \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p = \mathbb{Z}_p + \mathbb{Z}_p\omega$; note that this last equality holds for *every* p, because $\{1, \omega\}$ is a $\mathbb{Z}$-basis of $\mathcal{O}$.

Calderon [1] states the following. We keep the author's hypotheses and index names and set the display in our own notation.

> **Conjecture 6.5.** Let p satisfy (6.3), and suppose $1 \leq \gamma \leq \alpha \leq p - 1$ and $1 \leq \delta \leq \beta \leq p - 1$. Then

(2)
$$R_\omega(pA + \alpha, pB + \beta; pC + \gamma, pD + \delta)$$
$$\equiv R_\omega(A, B; C, D)\, R_\omega(\alpha, \beta; \gamma, \delta) \binom{\alpha}{\gamma}^D \binom{\beta}{\delta}^C \pmod{p\, \mathcal{O}_{\omega,p}}.$$

2020 *Mathematics Subject Classification.* Primary 11B65; Secondary 11A07, 11R11, 05A10.

Key words and phrases. Bailey congruence, Lucas' theorem, rectangular Gaussian binomial coefficient, quadratic order, inert prime, Frobenius.

Three clauses are inherited from the surrounding text rather than restated in that environment. The defining relation of ω asks only for $\omega^2 - T\omega + N = 0$ with $T, N \in \mathbb{Z}$ and $\Delta < 0$; the ring $\mathcal{O}_\omega := \mathbb{Z}[\omega]$ is called an order in an imaginary quadratic field, so *maximality is never assumed.* Hypothesis (6.3) is

$$p > 5, \qquad p \nmid \Delta, \qquad \left(\frac{\Delta}{p}\right) = -1. \tag{3}$$

The index ranges $A \geq C \geq 1$, $B \geq D \geq 1$ come from the definition (1).

What [1] *proves* is his Theorem 1.3, and only for $\omega = i$, since the coefficient there is defined over $\mathbb{Z}[i]$ alone; its hypothesis is that $p \equiv 3 \bmod 4$ be an odd prime. For $\omega = i$ one has $\Delta = -4$ and $\left(\frac{-4}{p}\right) = -1$ exactly when $p \equiv 3 \bmod 4$, so Theorem 1.3 is precisely the case $T = 0$, $N = 1$ of what follows.

Theorem 1. *Let p be an odd prime and $T, N \in \mathbb{Z}$ with*

$$p \nmid \Delta \qquad \textit{and} \qquad \left(\frac{\Delta}{p}\right) = -1. \tag{4}$$

Let $A \geq C \geq 1$, $B \geq D \geq 1$, and let $1 \leq \gamma \leq \alpha \leq p-1$, $1 \leq \delta \leq \beta \leq p-1$. Then all three rectangular coefficients occurring in (2) *lie in $\mathcal{O}_{\omega,p}$, and* (2) *holds. In particular Conjecture 6.5 of* [1] *is true, and it remains true when* (3) *is weakened to* (4).

2. The proof

Assume (4). Then $x^2 - Tx + N$ is irreducible over $\mathbb{F}_p$, so $\mathfrak{p} := p\,\mathcal{O}_{\omega,p}$ is a maximal ideal, $\mathcal{O}_{\omega,p}$ is a discrete valuation ring with uniformiser p and residue field

$$\mathcal{O}_{\omega,p}/\mathfrak{p} \cong \mathbb{F}_{p^2},$$

in which $\{1, \omega\}$ is an $\mathbb{F}_p$-basis. Write v for the valuation with $v(p) = 1$, extended to $\mathbb{Q}(\omega)^\times$, and write $\equiv$ for congruence modulo $\mathfrak{p}$. We use exactly three facts.

(i) For $a, b \in \mathbb{Z}$ not both zero, $v(a + b\omega) = \min(v_p(a), v_p(b))$.
Indeed, with $m = \min(v_p(a), v_p(b))$ write $a + b\omega = p^m(u' + v'\omega)$; then u', v' are not both divisible by p, so $u' + v'\omega \not\equiv 0$ because $\{1, \omega\}$ is an $\mathbb{F}_p$-basis of the residue field, i.e. $u' + v'\omega$ is a unit of $\mathcal{O}_{\omega,p}$.

(ii) $\omega^p \equiv \bar{\omega}$. The map $x \mapsto x^p$ is the nontrivial element of $\mathrm{Gal}(\mathbb{F}_{p^2}/\mathbb{F}_p)$, so it interchanges the two roots of $x^2 - Tx + N$.

(iii) For $x, y \in \mathbb{F}_p$,

$$\prod_{t \in \mathbb{F}_p} (x + t\omega) = c_1 x, \qquad \prod_{t \in \mathbb{F}_p} (t + y\omega) = c_2 y, \qquad c_1 = \frac{\omega - \bar{\omega}}{\omega}, \quad c_2 = \bar{\omega} - \omega,$$

and $c_1, c_2 \neq 0$. For the first, $\prod_{t \in \mathbb{F}_p}(X + t) = X^p - X$ gives, on putting $X = x/\omega$ and multiplying by ω^p,

$$\prod_{t \in \mathbb{F}_p} (x + t\omega) = x^p - \omega^{p-1} x = x - \frac{\bar{\omega}}{\omega} x = c_1 x$$

by (ii) and $x^p = x$; the second is $\prod_t (t + y\omega) = (y\omega)^p - y\omega = y(\bar{\omega} - \omega)$. Finally $(\omega - \bar{\omega})^2 = \Delta \neq 0$ in $\mathbb{F}_p$ by (4), and ω is a unit by (i), so c_1 and c_2 are nonzero. Note that neither constant is evaluated below.

Lemma 2 (integrality)**.** *For all $A \geq C \geq 1$ and $B \geq D \geq 1$ we have $v(R_\omega(A, B; C, D)) \geq 0$; that is, $R_\omega(A, B; C, D) \in \mathcal{O}_{\omega,p}$.*

Proof. By (i) and $\min(a, b) = \#\{j \geq 1 : p^j \mid \text{both}\}$,

$$v(F(\mathcal{I} \times \mathcal{J})) = \sum_{j \geq 1} M_{\mathcal{I}}(p^j)\, M_{\mathcal{J}}(p^j),$$

where $M_{\mathcal{I}}(q)$ is the number of multiples of q in the interval $\mathcal{I}$. An interval of length C contains at least $\lfloor C/q \rfloor$ multiples of q, and $[1, C]$ contains exactly $\lfloor C/q \rfloor$. Applying this to $\mathcal{I} = [A - C + 1, A]$ and $\mathcal{J} = [B - D + 1, B]$ gives $v(F([A - C + 1, A] \times [B - D + 1, B])) \geq v(F([1, C] \times [1, D]))$. □

Proof of Theorem 1. Write $A' = pA + \alpha$, $B' = pB + \beta$, $C' = pC + \gamma$, $D' = pD + \delta$, and put $\mathcal{I} = [A' - C' + 1, A']$, $\mathcal{J} = [B' - D' + 1, B']$, so that $|\mathcal{I}| = C'$ and $|\mathcal{J}| = D'$ and the left side of (2) is $F(\mathcal{I} \times \mathcal{J})/F([1, C'] \times [1, D'])$.

Step 1: split off the p-divisible factors, exactly. By (i), $p \mid (u + v\omega)$ if and only if $p \mid u$ and $p \mid v$. An integer pk lies in $\mathcal{I}$ if and only if $p(A - C) + (\alpha - \gamma + 1) \leq pk \leq pA + \alpha$; since $1 \leq \alpha - \gamma + 1 \leq p - 1$ and $0 \leq \alpha \leq p - 1$, this says exactly $A - C + 1 \leq k \leq A$, so $\mathcal{I}$ contains exactly the C multiples $\{pk : A - C + 1 \leq k \leq A\}$. Likewise $\mathcal{J}$ contains exactly $\{pl : B - D + 1 \leq l \leq B\}$, and, because $1 \leq \gamma \leq p - 1$ and $1 \leq \delta \leq p - 1$, the interval $[1, C']$ contains exactly $\{pk : 1 \leq k \leq C\}$ and $[1, D']$ exactly $\{pl : 1 \leq l \leq D\}$. Therefore

$$\begin{aligned} F(\mathcal{I} \times \mathcal{J}) &= p^{CD}\, F([A - C + 1, A] \times [B - D + 1, B]) \cdot \mathrm{Num}, \\ F([1, C'] \times [1, D']) &= p^{CD}\, F([1, C] \times [1, D]) \cdot \mathrm{Den}, \end{aligned}$$

where Num and Den are the products over those pairs (u, v) of the respective boxes that do *not* have both coordinates divisible by p. The powers p^{CD} cancel, so as an identity in $\mathbb{Q}(\omega)^\times$,

$$R_\omega(A', B'; C', D') = R_\omega(A, B; C, D) \cdot \frac{\mathrm{Num}}{\mathrm{Den}}, \tag{5}$$

and by (i) both Num and Den are units of $\mathcal{O}_{\omega,p}$. In particular

$$v(R_\omega(A', B'; C', D')) = v(R_\omega(A, B; C, D)) :$$

the p-power bookkeeping cancels uniformly, on (4) alone.

Step 2: a residue census. Modulo $\mathfrak{p}$ the factor $u + v\omega$ depends only on $(u \bmod p,\ v \bmod p)$, so

$$\mathrm{Num} \equiv \prod_{(r,s) \neq (0,0)} (r + s\omega)^{e(r) f(s)}, \qquad \mathrm{Den} \equiv \prod_{(r,s) \neq (0,0)} (r + s\omega)^{e_0(r) f_0(s)},$$

the products running over $(r, s) \in \mathbb{F}_p \times \mathbb{F}_p$, where $e(r)$ is the number of $u \in \mathcal{I}$ with $u \equiv r$ and $e_0,\ f,\ f_0$ are defined likewise for $[1, C'],\ \mathcal{J},\ [1, D']$. Since

$|\mathcal{I}| = pC + \gamma$ and $\mathcal{I}$ ends at $A' \equiv \alpha$,

$$e(r) = C + a_r, \quad a_r = [\, r \in S_\alpha \,], \qquad S_\alpha := \{\alpha - \gamma + 1, \dots, \alpha\},$$

and similarly $e_0(r) = C + a'_r$ with $a'_r = [\, r \in \{1, \dots, \gamma\} \,]$, $f(s) = D + b_s$ with $b_s = [\, s \in S_\beta \,]$, $S_\beta := \{\beta - \delta + 1, \dots, \beta\}$, and $f_0(s) = D + b'_s$ with $b'_s = [\, s \in \{1, \dots, \delta\} \,]$. *The residue* 0 *lies in none of the four sets* S_α, $\{1, \dots, \gamma\}$, S_β, $\{1, \dots, \delta\}$ — this is exactly what $1 \le \gamma \le \alpha \le p - 1$ and $1 \le \delta \le \beta \le p - 1$ buy — so $a_0 = a'_0 = b_0 = b'_0 = 0$, and the exponent of $(r + s\omega)$ in Num/Den is

$$e(r)f(s) - e_0(r)f_0(s) = C(b_s - b'_s) + D(a_r - a'_r) + (a_r b_s - a'_r b'_s).$$

Step 3: three groups. Summing the three terms separately, and using $a_0 = b_0 = 0$ to see that the first two extend over all s resp. all r,

$$\frac{\text{Num}}{\text{Den}} \equiv \left[\frac{\prod_{r \in S_\alpha} V(r)}{\prod_{r=1}^{\gamma} V(r)} \right]^D \cdot \left[\frac{\prod_{s \in S_\beta} W(s)}{\prod_{s=1}^{\delta} W(s)} \right]^C \cdot \frac{F(S_\alpha \times S_\beta)}{F([1, \gamma] \times [1, \delta])},$$

where $V(r) = \prod_{s \in \mathbb{F}_p} (r + s\omega)$ and $W(s) = \prod_{r \in \mathbb{F}_p} (r + s\omega)$.

Step 4: cardinality does the rest. The third group is $R_\omega(\alpha, \beta; \gamma, \delta)$ by the definition (1), since $S_\alpha = [\alpha - \gamma + 1, \alpha]$ and $S_\beta = [\beta - \delta + 1, \beta]$; it is a unit of $\mathcal{O}_{\omega,p}$ because every entry of both boxes lies in $[1, p-1]$. By (iii), $V(r) = c_1 r$ and $W(s) = c_2 s$, and $|S_\alpha| = \gamma = |\{1, \dots, \gamma\}|$, so c_1^γ cancels *by cardinality alone*, leaving

$$\frac{\prod_{r \in S_\alpha} V(r)}{\prod_{r=1}^{\gamma} V(r)} = \frac{\alpha(\alpha - 1) \cdots (\alpha - \gamma + 1)}{\gamma!} = \binom{\alpha}{\gamma} \quad \text{in } \mathbb{F}_p,$$

and symmetrically the second group contributes $\binom{\beta}{\delta}^C$. Hence, writing $U = \text{Num/Den}$ and $Z = R_\omega(\alpha, \beta; \gamma, \delta) \binom{\alpha}{\gamma}^D \binom{\beta}{\delta}^C$, we have $U \equiv Z$ with U, Z units, and by (5)

$$R_\omega(A', B'; C', D') - R_\omega(A, B; C, D)\, Z = R_\omega(A, B; C, D)\, (U - Z).$$

Here $v(U - Z) \ge 1$ and $v\big(R_\omega(A, B; C, D)\big) \ge 0$ by Lemma 2, so the left side has valuation at least 1, which is (2). Nothing in the argument used $\Delta < 0$, $p > 5$, the unit group of $\mathcal{O}$, or the maximality of $\mathcal{O}$. □

Remark. Two of the four sets in Step 2 exclude 0 only because $\gamma \le \alpha \le p - 1$. Drop the digit bounds and 0 enters S_α, at which point $V(0) = 0$ and Step 4 collapses.

Remark. At a split p fact (i) fails — p divides $u + v\omega$ for pairs with $p \nmid u$ — and the congruence fails in general: for $\omega = i$ and $p = 13$ it already fails at $(A, B, C, D) = (1, 2, 1, 1)$ with $\alpha = \beta = \gamma = \delta = 1$. Theorem 1 does *not* answer Problem 6.7 of [1], which asks for a *pair* of local congruences at the two primes above a split p.

Artifacts. Theorem 1 is a hand proof, and the program accompanying this note does not stand in for it. That program, `verify.py`, re-derives the finite combinatorial content of Steps 1, 2 and 4 and of Lemma 2 over stated finite ranges; checks facts (i)–(iii) in each family it uses; exhausts (2) itself, without failure, over the index ranges it prints, in families satisfying (4) both inside and outside (3); and reproduces the split-prime failure quoted in the remark above. It uses the Python standard library and exact integer arithmetic alone, and reports what it does not cover. A recorded run is included as `verify.output.txt`.

References

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